Agastya Mandalam

ML Model for Crop Demand Prediction

Research & Planning Document

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1. Objective

Build a farmer-facing ML model to predict crop demand and price trends, giving farmers (sellers) actionable insights: when to sell, where to sell, how much to produce, and which crop has rising demand. Output is a visual dashboard, not just a chatbot response.

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2. Data Sources

2.1 APIs (Preferred over static datasets)

APIs are preferred because they provide scalability, automation, and real-time data exchange, which is better than static CSVs for a production model.

- IMD API - Weather and rainfall data (imd.gov.in)
- Agmarknet API - Mandi/market prices across India
- eNAM API - National Agri Market real-time data
- data.gov.in - Crop production statistics
- DGCIS / dgciskol.nic.in - Import/export data (still to be sourced)

2.2 Datasets Already Collected

Dataset	Year Range	Rows	Contents	Gap
ICRISAT District Level	1966-2017	16,146	Area, production, yield per crop per district (80 columns, 20+ crops)	No prices, no weather
Crop Production Statistics	1997-2020	345,336	55 crops, 37 states, 6 seasons, area/production/yield	No prices, no weather

2.3 Additional Data Sources (from ICRISAT portal)

- Crop production by district: data.icrisat.org/dld/src/crops.html
- Irrigation data: data.icrisat.org/dld/src/irrigation.html
- Fertilizer/Input data: data.icrisat.org/dld/src/inputs.html
- Environmental data (soil, rainfall, landuse): data.icrisat.org/dld/src/biophysical.html
- Harvest prices: data.icrisat.org/dld/src/prices.html

3. Model Options & Progression

Stage	Model	Why
MVP	Multivariate LSTM	Handles sequential time-series data naturally. Captures lag effects (e.g., rain 2 months ago affecting yield today). Large datasets from Agmarknet + IMD justify the complexity over XGBoost.
V2	Temporal Fusion Transformer (TFT)	State-of-the-art for agri forecasting. Has built-in attention weights for interpretability - can visualize which features drove the prediction. Best for multi-horizon forecasting.
Seasonality Layer	+ Prophet	Added alongside LSTM/TFT to handle Indian seasonal cycles (Kharif, Rabi), festival demand spikes, and monsoon patterns.

Note on XGBoost: Initially considered but moved away from it for this use case because crop data is inherently sequential (daily/weekly price series), and LSTM naturally captures lag effects. XGBoost treats each row independently and does not model time dependencies. For large Agmarknet datasets, LSTM is the stronger choice.

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4. Model Features (What Goes Into the Model)

4.1 Core Features

Feature	Source	Why It Matters
Historical mandi prices	Agmarknet API	Core prediction target - weekly/daily modal price per crop per mandi
Mandi arrival quantity	Agmarknet API	Signals oversupply - rising arrivals = price drop incoming
Crop calendar / harvest season	ICRISAT + Agmarknet	Demand spike patterns tied to Kharif/Rabi seasons
Rainfall (lagged 2-3 months)	IMD API	Rainfall 2-3 months ago affects yield today - critical lag feature
MSP (Minimum Support Price)	data.gov.in	Policy-driven price floor - defines farmer's sell/hold decision
Festival/event calendar	Manual encoding	Demand spikes during Navratri, Diwali, harvest festivals
Import/Export balance	DGCIS	Sudden import policy changes can crash or spike prices
FPO aggregation volumes	Internal (Agastya)	Unique data point - how much produce our FPOs are holding
Cold storage availability	State agri dept	Affects farmer holding strategy and supply release timing

4.2 Features Often Neglected in GitHub Projects

- MSP changes (policy-driven price floors that directly affect farmer sell decisions)
- Interstate transport/logistics delays
- Cold storage capacity per region
- FPO aggregation volumes (unique ground-truth supply data)
- Rainfall as a lagged feature - most projects use current rainfall; correct approach is 2-3 month lag

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5. Feature Engineering on Time Series

This section covers how to process and engineer the raw datasets into training-ready features for the LSTM model.

5.1 Aligning Different Datasets

The two primary datasets (ICRISAT 1966-2017, Crop Stats 1997-2020) have different year ranges. Strategy:

1. Create a master reference CSV with Year, State, District, Crop as the join keys
2. Inner join both datasets on overlapping years (1997-2017 intersection) to get a clean 20-year window with all features
3. Preserve older ICRISAT data (1966-1996) separately for pre-training or long-term trend analysis

Structural alignment techniques to use:

- Melting and pivoting - reshape wide-format ICRISAT data (one column per crop) into long format (one row per crop) to match Crop Stats structure
- Changing data structure/format - normalize column names, unit conversions (1000 ha to ha, etc.)
- Handle temporal missing scales - fill gaps using forward-fill for slow-moving features (soil type), interpolation for price/weather

5.2 Time Series Feature Engineering Techniques

Technique	What It Does	When to Use
A. Crop-Commodity Features	Cross-crop correlation features. E.g., when rice price rises, does wheat demand shift? Capture substitution effects between crops.	When modelling demand shifts across crop categories
B. Volatility Windows	Rolling standard deviation over 4-week, 8-week, 12-week windows. Captures periods of price instability which affect farmer sell decisions.	As input to LSTM alongside price itself
C. Macro-Lags	Lagged versions of macroeconomic features (rainfall, import policy, MSP) over multiple periods. Captures delayed effects on price.	Rainfall (lag 60-90 days), MSP (lag 30 days), export ban (lag 7-14 days)

5.3 Encoding Cyclical & Seasonal Patterns

India's agricultural calendar has strong Kharif/Rabi seasonality and festival cycles. Standard month encoding (1-12) is wrong because it treats December and January as far apart. Use trigonometric encoding instead:

- Cyclical patterns (harvest years, multi-year cycles): Fourier transformation - decomposes the repeating signal into sine/cosine components
- Seasonal patterns (Kharif/Rabi, monsoon onset): Sine/cosine transformation of month or week-of-year
- Tenure features: Number of weeks/months since last season start, days to next MSP revision

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6. Master Feature Creation Guide

Based on the datasets collected and the India-specific agri context, the following derived/engineered features are recommended as high-signal inputs for the LSTM model:

Feature Name	How to Create	Signal It Carries
price_lag_1w / 4w / 8w	Shift mandi price column by 1, 4, 8 weeks	Short and medium-term momentum. Strong predictor for LSTM.
arrival_rolling_mean_4w	4-week rolling average of mandi arrival quantity	Supply buildup signal. Rising arrivals = price pressure incoming.
price_volatility_8w	Rolling std dev of price over 8 weeks	Market instability. High volatility = farmer holds back produce.
rain_lag_60d	IMD rainfall shifted 60 days back	Yield impact. Drought 2 months ago shows up in supply shortage now.
season_sin / season_cos	sin(2*pi*month/12) and cos(2*pi*month/12)	Cyclical seasonal encoding. Avoids discontinuity at year boundary.
kharif_flag / rabi_flag	Binary: 1 if current month is in Kharif (Jun-Nov) or Rabi (Nov-Apr)	Harvest season indicator. Prices typically fall at peak harvest.
msp_gap	Current mandi price minus MSP for that crop	How far above/below the floor price is. Negative = distress sale zone.
district_supply_ratio	District production / State production for that crop-year	Local supply concentration. High ratio = price discovery happens here.
crop_substitution_index	Correlation of price movement across 2-3 substitutable crops (e.g. wheat-rice)	Demand shift signal. When one price rises, demand moves to substitute.
fpo_holding_volume	Aggregated volume held by Agastya FPOs (internal data)	Unique feature. Tells the model how much supply is being withheld.

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7. Model Output (Dashboard for Farmers)

The model output is NOT a chatbot answer. It powers a visual farmer analytics dashboard:

4. Price forecast curve - predicted price for the next 4-12 weeks
5. Filters - by crop type, region/mandi, season
6. Multiple curves on one chart - predicted price vs MSP vs last year same period
7. Heatmap of demand by region - which mandis are showing rising demand
8. Best time to sell indicator - a simple recommendation derived from the forecast

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8. Research Paper Reference

Reference: 'Analysis of Demand Forecasting of Agriculture using Machine Learning Algorithm' - Chelliah, Latchoumi, Senthilselvi (2024), Environment Development and Sustainability, Springer.

Key takeaways from this paper:

- Used 6 factor inputs: land relief, water availability, temperature, soil characteristics, import/export data, GPS/satellite imagery
- Architecture: CNN for satellite image features (drought/NDVI detection) + XGBoost for tabular data - combined into a 4-layer pipeline (Ingestion, Transformation, Training, Inference)
- Achieved ~96% accuracy on drought classification using CNN on 35 satellite images from Kolar district
- For our use case: skip the CNN satellite layer for MVP. Use IMD rainfall as a proxy. Add satellite features in V2.
- Their use of a bottom-up data collection approach (district -> state -> national) is directly relevant to our FPO structure

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9. All Data Sources & Links

Government / Official APIs

- IMD Weather API: https://mausam.imd.gov.in/
- Agmarknet (Mandi Prices): https://agmarknet.gov.in/
- eNAM (National Agri Market): https://www.enam.gov.in/
- data.gov.in (Crop Production Stats): https://data.gov.in/
- DGCIS (Import/Export): https://dgciskol.nic.in/

ICRISAT Data Portal (1966-2018)

- Crop Production: http://data.icrisat.org/dld/src/crops.html
- Irrigation Data: http://data.icrisat.org/dld/src/irrigation.html
- Fertilizer/Inputs: http://data.icrisat.org/dld/src/inputs.html
- Environmental Data (soil, rainfall, landuse): http://data.icrisat.org/dld/src/biophysical.html
- Harvest Prices: http://data.icrisat.org/dld/src/prices.html

Datasets Collected

- ICRISAT District Level Data CSV (1966-2017, 16k rows, 80 columns)
- Crop Production Statistics CSV (1997-2020, 345k rows, 55 crops)

Research Reference

- Springer Paper DOI: https://doi.org/10.1007/s10668-022-02783-9
